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Ducted fan aircraft propulsion system

Abstract

The disclosure relates to a ducted fan aircraft propulsion system and to an aircraft incorporating such a propulsion system. Example embodiments include a ducted fan aircraft propulsion system (300), including: a duct (301); a central body portion (302) having first and second ends (303, 304) and extending through the duct (301); a payload portion (305) extending from the first end (303) of the central body portion (302); and a rotor (306) extending across an internal volume (307) of the duct (301) from the central body portion (302), wherein the duct (301) includes an outwardly flared inlet end (308) such that an inlet air flow passage (309) between the central body portion (302) and an inner surface (310) of the duct (301) has a sectional area that decreases from the inlet end (308) of the duct (301) to the rotor (306).

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Background/Summary

(1) This specification is based upon and claims the benefit of priority from UK Patent Application Number 2313173.3 filed on 30 Aug. 2023, the entire contents of which are incorporated herein by reference.

BACKGROUND

Technical Field

(2) The disclosure relates to a ducted fan aircraft propulsion system and to an aircraft incorporating such a propulsion system.

Background of the Disclosure

(3) Certain types of vertical take-off and landing (VTOL) aircraft may use one or more ducted fans for lift and propulsion. A ducted fan is a thrust-generating mechanical fan or propellor that is mounted within a cylindrical duct or shroud. An example of an aircraft with a ducted fan is a coleopter, which uses a ducted fan as the primary fuselage of the aircraft. Multiple ducted fans may be employed in manned or unmanned aircraft. Advantages of ducted fans include lower noise, improved efficiency (due to reduced rotor tip losses) and a reduced rotor size. This may, however, lead to aircraft with a high disk loading (i.e. the ratio of weight of an aircraft to the total main rotor disc area) or a reduced payload capability and consequent higher power consumption during hover.

(4) FIG. 1 illustrates schematically a relationship between hover lift efficiency in terms of gross weight per unit power (in kg/kW) as a function of disc loading in terms of gross weight per unit thrust area (in kg/m.sup.2). Helicopters have a relatively low disc loading and a high hover lift efficiency, while VTOL aircraft have a relatively high disc loading and a low hover lift efficiency. Other aircraft such as tilt rotor, tilt wing and lift-fan aircraft lie between these extremes.

(5) FIG. 2 illustrates an example straight duct design for a ducted fan. The duct **200** has a circular cross section and comprises an inner surface **201** defining a uniform cross-sectional area along a longitudinal axis **202** of the duct **200**. An outer surface **203** of the duct has a uniform diameter along a first portion **204** of the duct **200** and a tapering diameter along a second portion **205** of the duct **200** towards an outlet end **206**. A rotor (not shown) provided within the interior of the duct forces airflow through the duct **200** from an inlet **207** to the outlet **206**, providing thrust for an aircraft.

(6) A general problem to be addressed is how to improve the hover lift efficiency of a ducted fan propulsion system.

SUMMARY

(7) According to a first aspect there is provided a ducted fan aircraft propulsion system, comprising: a duct; a central body portion having first and second ends and extending through the duct; a payload portion extending from the first end of the central body portion; and a rotor extending from the central body portion within the duct, wherein the duct comprises an outwardly

- flared inlet end such that an inlet air flow passage between the central body portion and an inner surface of the duct has a sectional area that decreases from the inlet end of the duct to the rotor.
- (8) The reducing air flow passage encourages a larger volume of air into the duct, which aids lift due to the flared inlet end of the annular wing, thereby improving the hover lift efficiency of the aircraft.
- (9) The duct may comprise an outwardly flared outlet end such that an outlet air flow passage between the central body portion and the inner surface of the duct has a sectional area that increases from the rotor to the outlet end of the duct. The increase in sectional area from the rotor to the outlet end of the duct may be smaller than the decrease in sectional area from the inlet end to the rotor.
- (10) The duct may form an annular wing having a decreasing cross-section from the inlet end of the duct to the outlet end of the duct.
- (11) The rotor may be positioned within the duct extending across a minimum of the sectional area of the duct.
- (12) An upper part of an outer surface of the payload portion may define a spheroidal shape. The spheroidal shape may for example be an oblate spheroid having a shorter axis along a longitudinal axis of the central body portion.
- (13) A lower part of the outer surface of the payload portion may taper from the spheroidal shape to the central body portion.
- (14) The central body portion may comprise a motor connected to drive the rotor via a rotor shaft. The motor may be electric. The system may comprise an electric storage unit and an electric power controller configured to provide electric power to the motor.
- (15) According to a second aspect there is provided an aircraft comprising a ducted fan propulsion system according to the first aspect.
- (16) The aircraft may comprise a single ducted fan propulsion system.
- (17) The duct may form an outer fuselage of the aircraft.
- (18) The aircraft may be a coleopter.
- (19) The skilled person will appreciate that except where mutually exclusive, a feature described in relation to any one of the above aspects may be applied mutatis mutandis to any other aspect. Furthermore except where mutually exclusive any feature described herein may be applied to any aspect and/or combined with any other feature described herein.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments will now be described by way of example only with reference to the accompanying drawings, which are purely schematic and not to scale, and in which:
- (2) FIG. 1 is a chart illustrating hover lift efficiency as a function of disc loading (derived from https://www.wikiwand.com/en/Disk_loading);
- (3) FIG. 2 is a series of line drawings illustrating an example straight duct for a conventional ducted fan;
- (4) FIG. 3 is a line drawing of an example design for a ducted fan aircraft propulsion system;
- (5) FIG. 4 is a cross-sectional drawing of the example ducted fan aircraft propulsion system of FIG. 3;
- (6) FIG. 5a is a line drawing of a top view of the example ducted fan aircraft propulsion system of FIG. 3;
- (7) FIG. 5b is a line drawing of a side view of the example ducted fan aircraft propulsion system of FIG. 3;
- (8) FIG. 6 is a schematic drawing of airflow around and through an example ducted fan;

- (9) FIG. 7 is an output airflow plot from a computational fluid dynamics model of an example ducted fan;
- (10) FIG. 8 is an output static pressure plot from a computational fluid dynamics model of an example ducted fan;
- (11) FIG. 9 is an output velocity streamline plot from a computational fluid dynamics model of an example ducted fan; and
- (12) FIG. 10 is a schematic diagram of an example aircraft comprising a ducted fan propulsion system.

DETAILED DESCRIPTION

- (13) FIGS. 3, 4, 5a and 5b illustrate an example design for a ducted fan propulsion system **300**. The system **300** comprises a duct **301** surrounding a central body portion **302**. The central body portion **302** extends through the duct **301** from a first end **303** to a second end **304**. A payload portion **305** extends from the first end **303** of the central body portion **302**. A rotor **306** extends across an internal volume **307** of the duct from the central body portion **302**. The system **300** is rotationally symmetric about a central axis **311** extending through the central body portion **302**.
- (14) The duct **301** comprises an outwardly flared inlet end **308** such that an annular inlet air flow passage **309** between an outer surface of the central body portion **302** and an inner surface **310** of the duct **301** has a sectional area that decreases from the inlet end **308** of the duct **301** to the rotor **306**.
- (15) In the example system **300** in FIG. 3, the duct **301** also comprises an outwardly flared outlet end **312** such that an outlet air passage **313** between the central body portion **302** and the inner surface **310** of the duct **301** has a sectional area that increases from the rotor **306** to the outlet end **312**.
- (16) The cross-sectional shape of the duct **301** forms an annular wing having a decreasing cross-section from the inlet end **308** of the duct **301** to the outlet end **312** of the duct **301**. The annular wing shape enhances lift of the propulsion system **300** due to airflow through the duct **301**. More thrust can thereby be achieved at lower power values, leading to improved hover efficiency.
- (17) The rotor **306** is positioned in the middle of the duct **301** at a position where the sectional area of the duct **301** is at a minimum.
- (18) The payload portion **305** in FIG. 3 has an upper portion with an outer surface that defines a spheroidal shape. The spheroidal shape in the illustrated example is an oblate spheroid having a shorter axis along the longitudinal axis **311** of the central body portion **302**. The payload portion **305** may take other shapes, for example a spherical shape or a prolate spheroidal shape.
- (19) FIG. 4 illustrates a sectional view of the system **300**, which is taken along section A-A indicated in FIG. 3. FIGS. 5a and 5b illustrate a top view and a side view respectively of the system **300**.
- (20) FIG. 6 is a schematic diagram illustrating how the various geometries of the system **300** increase lift and reduce the work done by the rotor or fan **306**. Firstly, the outwardly flared inlet **308** encourages a larger volume **601** of air into the fan **306**, with a reducing sectional area between the inlet **308** and the fan **306** having reduced obstruction to airflow. Secondly, the shape of the payload body **305** forms an accelerated flow of air **602** over the payload body **305** and into the duct inlet **603**. The shape of the payload body **305** in combination with the outwardly flared shape of the duct inlet end forms a smooth and compressing flow of air through the duct inlet **603**. The outwardly flared inlet end **308** maintains a high-pressure volume **604** under annular wing shape of the duct inlet **308**, which increases suction over the wing. A low-pressure region **605** is created within the duct between the fan **306** and the duct outlet **606**, which creates a high pressure underneath the duct body with expansion at the duct outlet **606**.
- (21) Airflow **607** around the outer surface of the duct is directed to the inlet **603** and speeded up over the annular wing of the duct. Airflow **608** further away from the duct is not blocked by the annular wing.

- (22) The general principles illustrated in FIG. 6 and described above are shown in more detail in the plot of FIG. 7, which illustrates results from an Ansys Fluent computational fluid dynamics (CFD) model of the duct and central body portion with the payload portion. The left half of the plot illustrates static pressure, while the right half illustrates velocity streamlines. These are illustrated separately, with associated pressure values and velocity magnitudes, in FIGS. 8 and 9.
- (23) Based on CFD simulations such as those illustrated in FIGS. 7-9, it was calculated that the duct design can improve lift in hover by around 67% compared with a simple straight duct design, i.e. the duct design of the type illustrated in FIG. 2. This is mirrored in physical prototyping test results on a 1:33 scale model that measured both lift and power of the system, in which a 65.5% improvement in lift efficiency (overall lift/power) was achieved over a simple straight duct design. When the vehicle is experiencing ground effect, a 49% increase in lift efficiency compared to a straight duct design was seen in prototype physical testing.
- (24) Considering the trends and results from both CFD and physical prototyping, along with benchmarking of existing products, it is estimated that a vehicle based on the duct design could achieve a hover lift efficiency value of around 2.92 at a disk loading of 439 kgm.sup.-2 for a 3.1 tonne maximum take-off weight. This would place such an aircraft in a region similar to a tilt wing aircraft in the schematic plot of FIG. 1.
- (25) FIG. 10 illustrates schematically an example aircraft **1000** with a propulsion system based on the duct design described above. The aircraft **1000** comprises the propulsion system as described above in relation to FIGS. 3-5. Also illustrated schematically in FIG. 10 is a motor **1001** connected to drive a rotor shaft **1004** connected to the rotor **306**. The motor **1001** may be an electric motor, which is powered from an electric storage unit **1003** with an electric power controller **1002**. The electric storage unit **1003** may comprise a battery module. The electric power controller **1002** may function to control electric power provided to the motor **1002** and to maintain a required voltage supply using a power converter module that is provided with a DC voltage supply from the storage unit **1003**. The electric storage unit **1003**, controller **1002** and electric motor **1001** may be provided in the payload portion **305**, in the central body portion **302** or distributed between the payload portion **305** and the central body portion **302**.
- (26) The central body portion **302** and the duct **301** are connected to each other by a plurality of struts **1004** extending between the central body portion **302** and the inner surface **310** of the duct **301**. The struts **1004** are illustrated in FIG. 10 as extending across an inlet end of the duct **301**, but may be provided elsewhere, for example towards the outlet end or closer to the rotor **306** on either the inlet or outlet side.
- (27) In the example aircraft **1000** in FIG. 10, a single ducted propulsion system forms the aircraft **1000**, making the aircraft take the form of a coleopter. In alternative examples the aircraft may comprise a plurality of ducted fan propulsion systems linked together. Linking multiple systems together may improve stability of the aircraft, particularly during hover.
- (28) In an example design, the rotor or fan **306** of the aircraft **1000** may be around 3 m in diameter. The fan may have a maximum speed of 2000 RPM and with a power consumption of around 1100 kW. The duct **301** may have a maximum outer diameter of around 6 m. The shape of the duct **301** is designed in conjunction with the payload body **305** to increase the vehicle lift efficiency, as described above. In a general aspect, an outer diameter of the rotor may be between around 2 and 4 metres and a maximum outer diameter of the duct may be between around 5 and 10 metres. A total weight of the aircraft may be between around 2 and 10 tonnes.
- (29) Other embodiments are within the scope of the invention, which is defined by the appended claims.

Claims

1. A ducted fan aircraft propulsion system, comprising: a duct; a central body portion having first and second ends and extending through the duct; a payload portion extending from the first end of the central body portion, wherein an upper part of an outer surface of the payload portion defines a spheroidal shape; and a rotor extending across an internal volume of the duct from the central body portion, wherein the duct comprises an outwardly flared inlet end such that an inlet air flow passage between the central body portion and an inner surface of the duct has a sectional area that decreases from the inlet end of the duct to the rotor, and wherein a cross-sectional shape of the duct forms an annular wing having a decreasing cross-section from the inlet end of the duct to an outlet end of the duct.
 2. The ducted fan aircraft propulsion system of claim 1, wherein the duct comprises an outwardly flared outlet end such that an outlet air flow passage between the central body portion and the inner surface of the duct has a sectional area that increases from the rotor to the outlet end of the duct.
 3. The ducted fan aircraft propulsion system of claim 1, wherein the rotor is positioned within the duct extending across a minimum of the sectional area of the duct.
 4. The ducted fan aircraft propulsion system of claim 1, wherein the spheroidal shape is an oblate spheroid having a shorter axis along a longitudinal axis of the central body portion.
 5. The ducted fan aircraft propulsion system of claim 1, wherein a lower part of the outer surface of the payload portion tapers from the spheroidal shape to the central body portion.
 6. The ducted fan aircraft propulsion system of claim 1, comprising a motor connected to drive the rotor via a rotor shaft.
 7. The ducted fan aircraft propulsion system of claim 6, wherein the motor is an electric motor.
 8. The ducted fan aircraft propulsion system of claim 7, comprising an electric storage unit and an electric power controller configured to provide electric power to the motor.
 9. An aircraft comprising the ducted fan aircraft propulsion system according to claim 1.
 10. The aircraft of claim 9, wherein the aircraft comprises a single one of the ducted fan aircraft propulsion systems.
 11. The aircraft of claim 9, wherein the duct forms an outer fuselage of the aircraft.
 12. The aircraft of claim 11, wherein the aircraft is a coleopter.
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